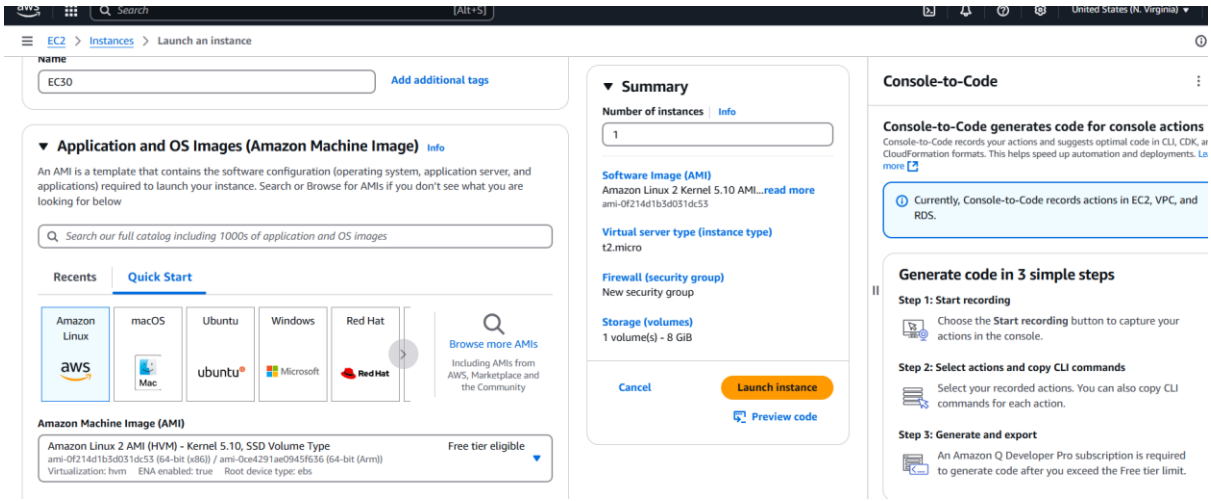
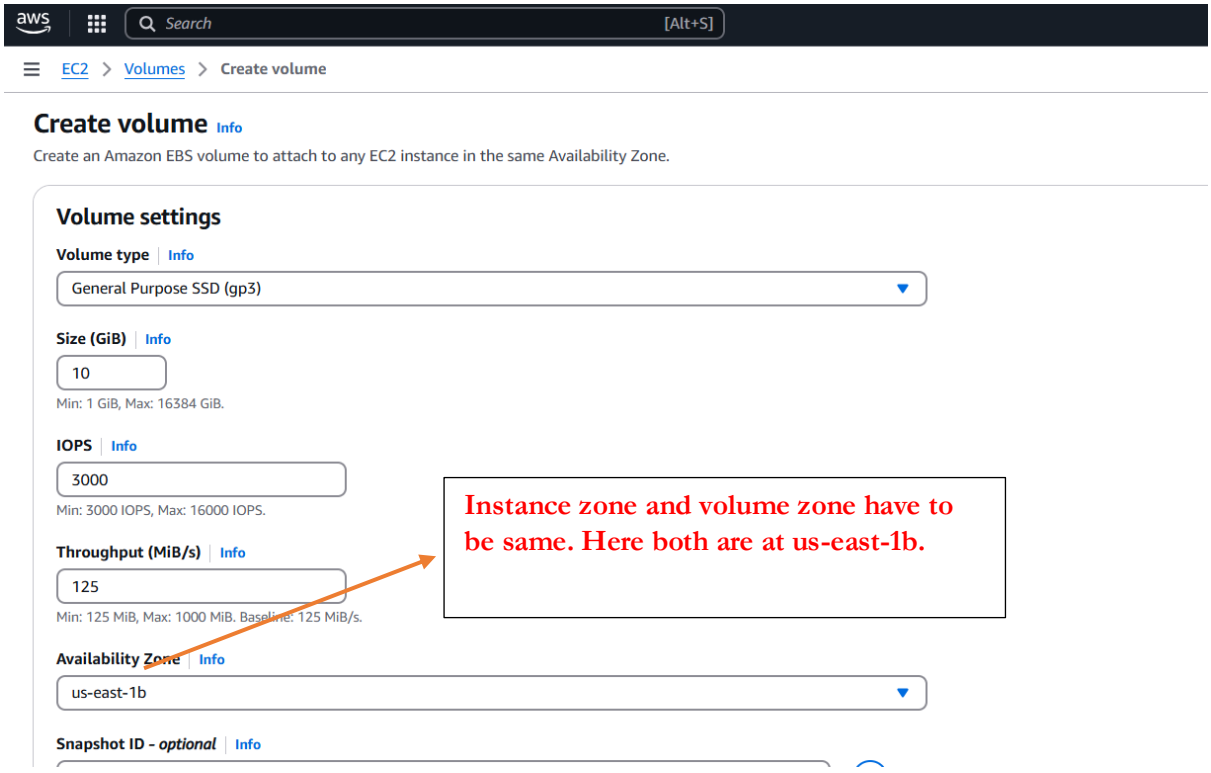
Task 3: Attach a volume to the EC2 instance in Linux

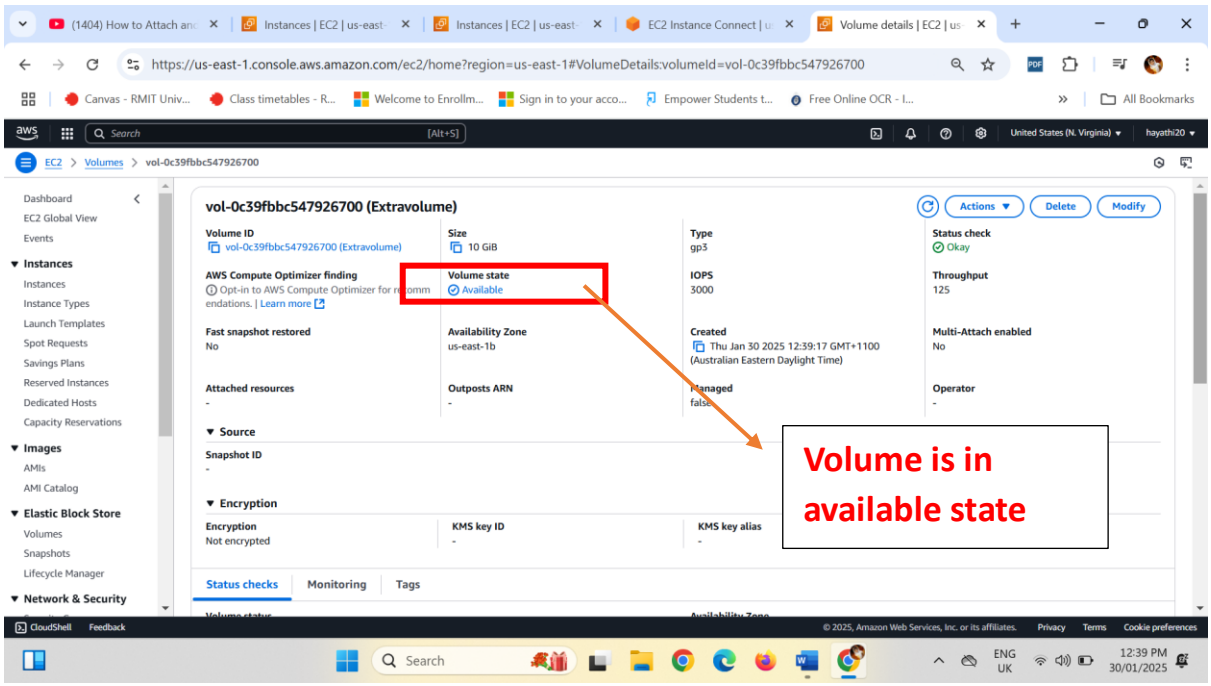
Step 1: Launching a new Instance



Step 2 : To Create a volume Goto-> Elastic Block store-> Volume

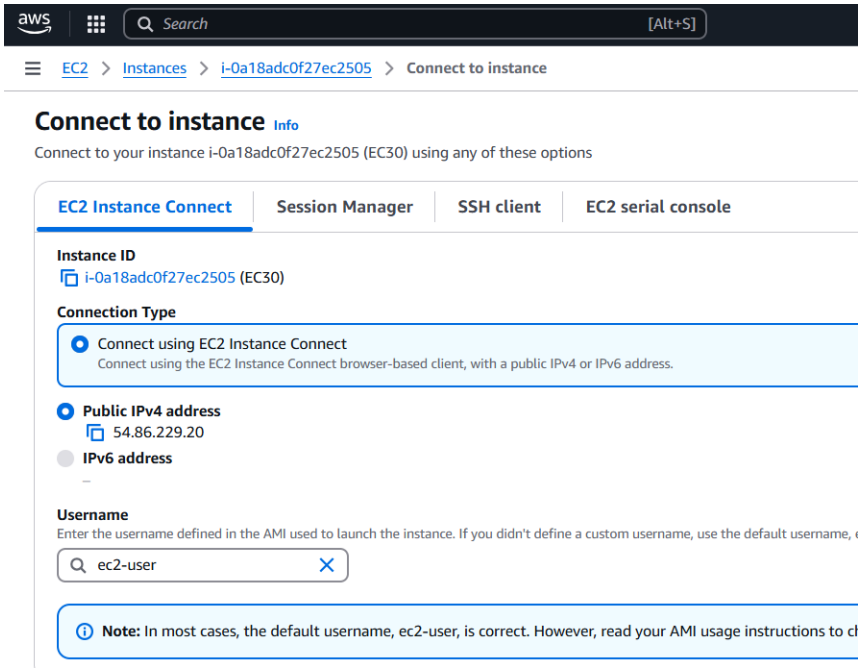


Step 3: Click on Create volume and a new volume is created.

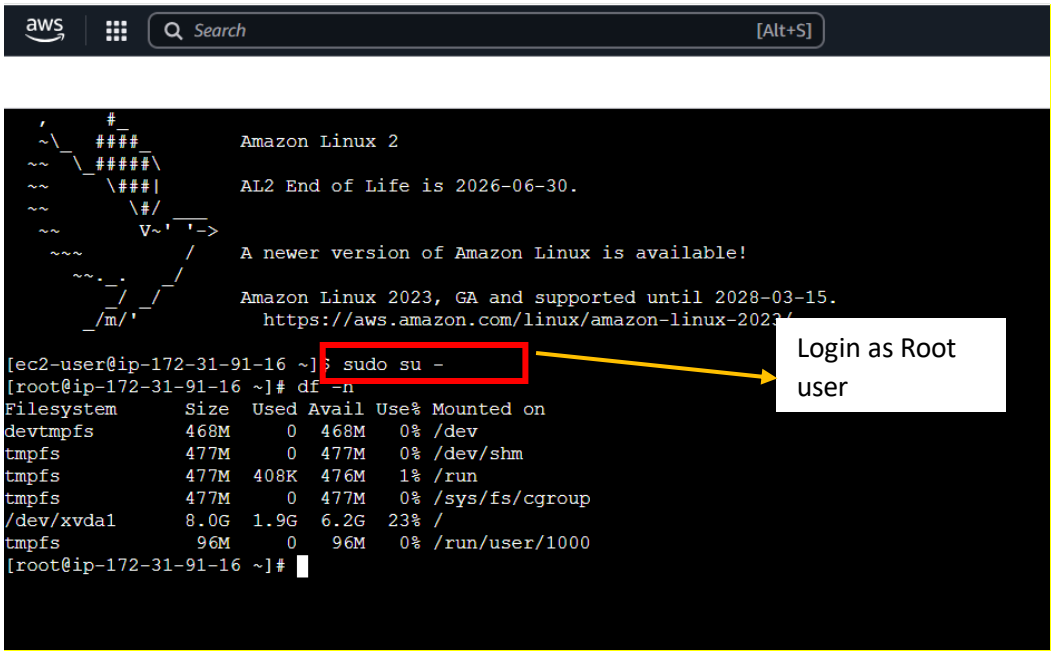


The next step is to connect this available volume with the Instance.

Step 4 : Connecting to EC2 Instance



After successful connection, Amazon Linux is opened.

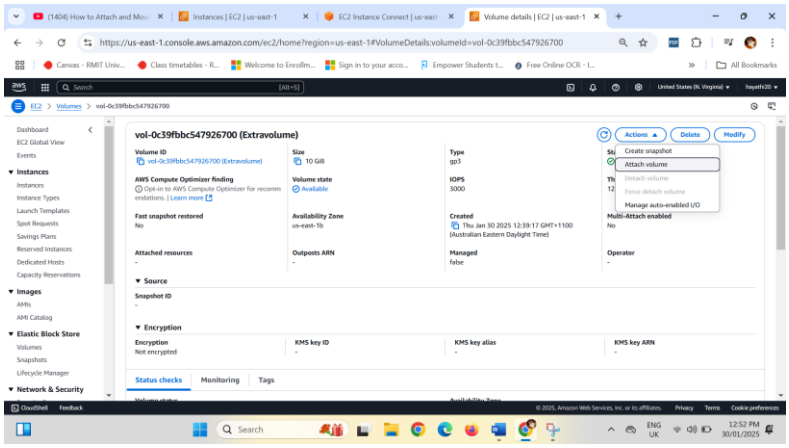


Use **df -h** command to check available volume connected with our instance.

```
[root@ip-172-31-91-16 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  408K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
[root@ip-172-31-91-16 ~]#
```

Step 5 : Attach the volume with the Instance

Goto Volume-> Select Attach volume



Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-0c39fbbc547926700 (Extravolume)

Connected with the available instance

Availability Zone

us-east-1b

Instance Info

i-06e7e2888668dc247
(EC30) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name Info

Select a device name

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

Now the attached volume state changed from Available ---> In-use

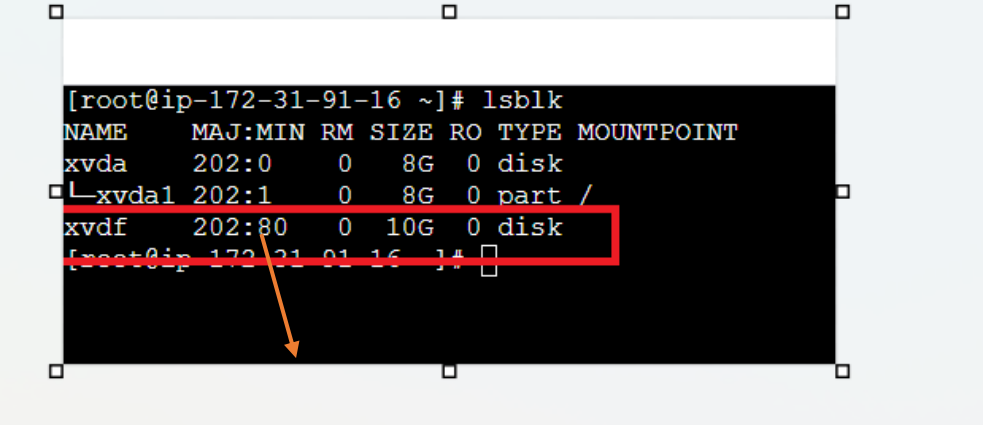
125	-	2025/01/30 12:39 GMT+11	us-east-1b	In-use	No alarms	+	i-06

Step 6: Then , Checking in EC2 Instance if it is showing under the EC30 Instance

```
[root@ip-172-31-91-16 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  412K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
[root@ip-172-31-91-16 ~]#
```

It is still not displayed here because we need to create a directory and mount the EBS volume to the Instance.

Step 7: Use the lsblk command to view your available disk devices and their mount points



Attached volume can now viewed in the Instance (EC30)

Step 8: Check if there is any file system on new EBS Volume:

file -s /dev/xvdf (If you see "Data", meaning you need to setup file system for this block device.)

```
xvdf    202:80   0  10G  0 disk
[root@ip-172-31-91-16 ~]# file -s /dev/xvdf
/dev/xvdf: data
[root@ip-172-31-91-16 ~]#
```

Setting up file system:

1. Create a file system on volume to mount it to EC2:

mkfs -t xfs /dev/xvdf

```
[root@ip-172-31-91-16 ~]# mkfs -t xfs /dev/xvdf
meta-data=/dev/xvdf              isize=512    agcount=4, agsize=655360 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=0 inobtcount=0
data     =                       bsize=4096   blocks=2621440, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]#
```

2. Create new directory:
   mkdir -p /apps/my-data/apps/volume/new-volume
   cd /apps/my-data/

```
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]#
```



Directory is created.

Step 9: Mount the volume

mount /dev/xvdf /apps/my-data/apps/volume/new-volume

```
[root@ip-172-31-91-16 ~]# lsblk
NAME    MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
xvda    202:0    0   8G  0 disk
└─xvda1 202:1    0   8G  0 part /
xvdf    202:80   0  10G  0 disk
[root@ip-172-31-91-16 ~]# file -s /dev/xvdf
/dev/xvdf: data
[root@ip-172-31-91-16 ~]# mkfs -t xfs /dev/xvdf
meta-data=/dev/xvdf              isize=512    agcount=4, agsize=655360 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=0 inobtcount=0
data     =                       bsize=4096   blocks=2621440, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]# mount /dev/xvdf /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 my-data]#
```

Step 10: To check if the mounting is successful, use df -h command

```
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]# mount /dev/xvdf /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 my-data]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  468K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
tmpfs            96M     0   96M   0% /run/user/0
/dev/xvdf        10G  104M  9.9G   2% /apps/my-data/apps/volume/new-volume
```

Mounting is successful and it is available in Instance (EC30)

```
[root@ip-172-31-91-16 my-data]# lsblk
NAME    MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
xvda    202:0    0   8G  0 disk
└─xvda1 202:1    0   8G  0 part /
xvdf    202:80   0  10G  0 disk /apps/my-data/apps/volume/new-volume
```

Previously it was not listed, and it is available now after successful mounting.

Also, it could be confirmed by viewing Storage in Particular Instance in AWS console.

